

Numerical Analysis PhD Exam, May 2014

You have 2 hours to answer the following questions. You must show all your work as neatly and clearly as possible and indicate the final answer clearly. You may not use a calculator.

1. Show that the fixed point equation $x = f(x)$ with a fixed point α can be solved iteratively, if $f'(\alpha) \neq 1$, by one of the following fixed point iteration formulas:

(a) $x_{n+1} = f(x_n)$

(b) $x_{n+1} = f^{-1}(x_n)$. Hint: The following formula for the derivative of an inverse is valid

$$[f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}.$$

(c) $x_{n+1} = (x_n + f(x_n))/2$

2. Let $f(x) \in C^1[a, b]$. Let $p(x)$ be a polynomial for which

$$\|f' - p\|_{\infty} \leq \epsilon$$

and define

$$q(x) = f(a) + \int_a^x p(t) dt, \quad a \leq x \leq b.$$

Show that $q(x)$ is a polynomial that satisfies

$$\|f - q\|_{\infty} \leq \epsilon(b - a).$$

3. Consider the integral

$$\int_0^{\pi} x^2 \cos x dx. \tag{1}$$

- (a) Consider a quadrature rule of the form

$$\int_0^{\pi} x^{\alpha} f(x) dx \approx Af(0) + B \int_0^{\pi} f(x) dx$$

where $\alpha > -1$, $\alpha \neq 0$ is a parameter. Determine the constants A, B so that the quadrature formula has degree of exactness one.

- (b) Use the formula in part (a) to approximate the integral (1).

4. For the function $f(x) = \ln(1 + x)$ for $x \in [0, 1]$, find the minimax approximation polynomial of degree one. Give the exact value of the minimax error.

5. Assume that you are solving the initial value problem

$$\begin{aligned} y' &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= \alpha. \end{aligned}$$

- (a) Derive the formula for the global error of the numerical solutions for the ODE problem above obtained via Euler's method. Hint: The formula is $M = \|Y''\|_\infty$:

$$|Y(t_i) - w_i| < \frac{hM}{2L} [e^{L(b-a)} - 1] .$$

- (b) Compute the value of $M = \|Y''\|_\infty$ necessary to apply the global error formula above to the specific ODE problem

$$\begin{aligned} y' &= \sin(t + 2y) + e^t, \quad 0 \leq t \leq 1, \\ y(0) &= 0. \end{aligned}$$